Theory of Computation $\qquad \qquad \qquad 1-7 \qquad \qquad \qquad$ Review of Mathematical Theory

Let us discuss each one by one -

i) Negation

This proposition is also called NOT proposition. It is denoted by $\neg$ or $\sim$ or having horizontal bar on letter.

If S is a proposition then $\neg S$ defines NOT S. The truth table for this connective is

S	$\neg S$
T	F
F	T

ii) Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by $A \land B$. The truth table for this connective is

A	B	$A \land B$
T	T	T
T	F	F
F	T	F
F	F	F

iii) Disjunction (OR)

If A and B are two propositions then disjunction of A and B is denoted by $A \lor B$. The truth table for this connective is

A	B	$A \lor B$
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted by $A \to B$ or $A \implies B$.